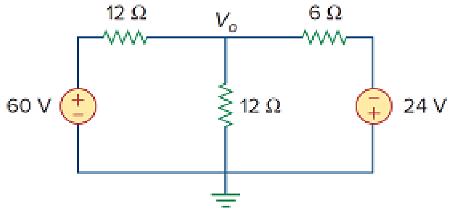
Chapter 3, Problem 11P

Problem

Find *Vo* and the power dissipated in all the resistors in the circuit of Fig. 3.60.

Figure 3.60:



Step-by-step solution

Step 1 of 5

Draw the following circuit diagram shown in Figure 1.

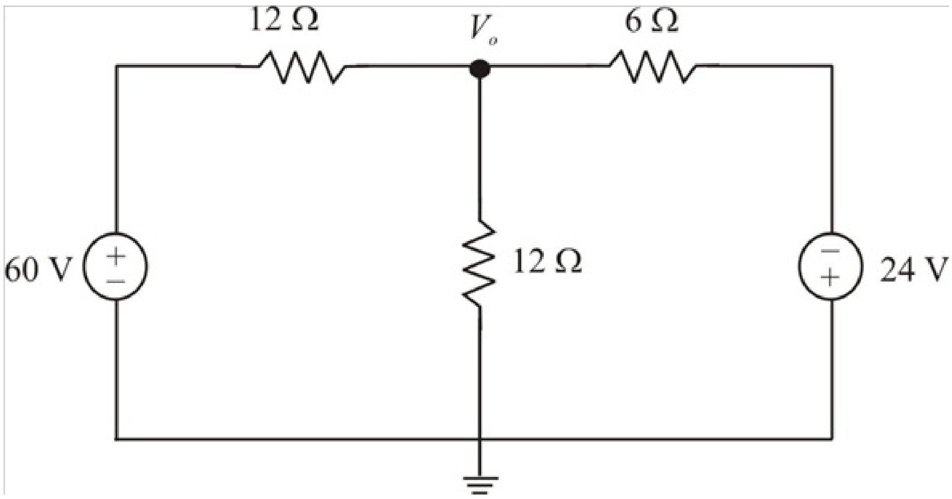


Figure 1

Step 2 of 5

The circuit with currents labeled is shown in Figure 2.

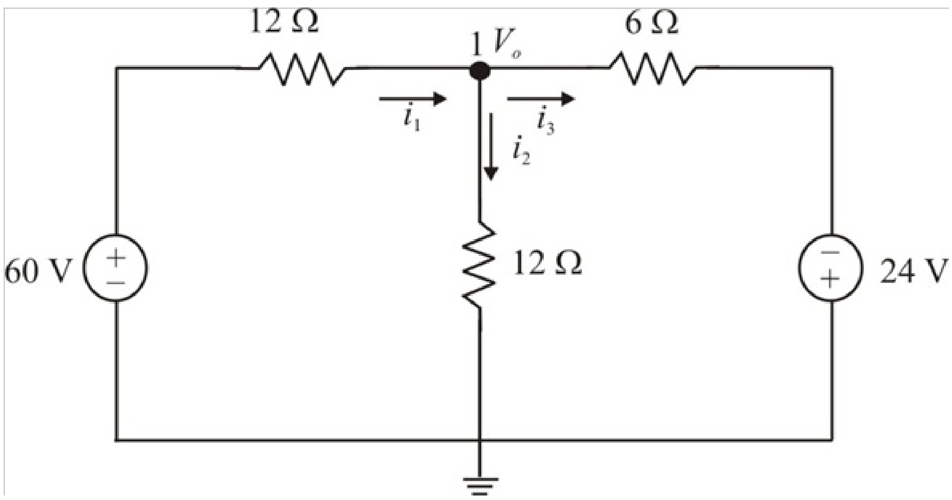


Figure 2

Step 3 of 5

Applying Kirchhoff's current law, the sum of currents entering the node is equal to the sum of the currents leaving from that node.

For the node 1, Kirchhoff's current law gives,

$$i_1 = i_2 + i_3$$
$$\frac{60 - V_o}{12} = \frac{V_o}{12} + \frac{V_o + 24}{6}$$

Multiplying by 12 on both sides, we get

$$60 - V_o = V_o + 2V_o + 48$$
$$4V_o = 12$$
$$V_o = \frac{12}{4}$$
$$= 3\,\text{V}$$

Therefore the voltage V_o is $\boxed{3\,\text{V}}$.

Step 4 of 5

The current i_1 can be calculated as:

$$i_1 = \frac{60 - V_o}{12}$$
$$= \frac{60 - 3}{12}$$
$$= \frac{57}{12}$$
$$= 4.75\,\text{A}$$

The power dissipated in the first 12 ? resistor can be calculated as:

$$p_1 = i_1^2 R$$
$$= (4.75)^2 (12)$$
$$= 270.75\,\text{W}$$

Therefore the power dissipated in the first 12 ? resistor is $\boxed{270.75\,\text{W}}$.

Step 5 of 5

The current i_2 can be calculated as:

$$\begin{aligned} i_2 &= \frac{V_o}{12} \\ &= \frac{3}{12} \\ &= 0.25\,\text{A} \end{aligned}$$

The power dissipated in the middle 12 ? resistor can be calculated as:

$$\begin{aligned} p_2 &= i_2^2 R \\ &= (0.25)^2 (12) \\ &= 750\,\text{mW} \end{aligned}$$

Therefore the power dissipated in the middle 12 ? resistor is $\boxed{750\,\text{mW}}$.

The current i_3 can be calculated as:

$$\begin{aligned} i_3 &= \frac{V_o + 24}{6} \\ &= \frac{3 + 24}{6} \\ &= \frac{27}{6} \\ &= 4.5\,\text{A} \end{aligned}$$

The power dissipated in the 6 ? resistor can be calculated as:

$$\begin{aligned} p_3 &= i_3^2 R \\ &= (4.5)^2 (6) \\ &= 121.5\,\text{W} \end{aligned}$$

Therefore the power dissipated in the 6 ? resistor is $\boxed{121.5\,\text{W}}$.